

PVT

掌锐电子



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问题概述

- 问题背景：

- 【徐水】【耐久测试】【EC15S】TBOX未登录TSP



- 问题现象：

- 车辆已开通长连接功能，25日晚19:37分后无后台数据上报；无登出消息，无长连接浅休眠消息上报。车辆在徐水耐久三班测试，一直在上电使用；车机公网未发现异常，天气应用正常，GPS正常。

问题分析

5W分析:

1W: 为什么TBOX未登录TSP? 因为APP未重新进行私网APN拨号

2W: 为什么APP未重新进行私网APN拨号? 因为APP没有收到服务mtktelephony服务回调APN断开状态

3W: 为什么APP没有收到底层服务mtktelephony服务回调APN断开状态? 因为服务mtktelephony服务没有回调上报APN断开状态

4W: 为什么服务mtktelephony服务没有回调上报? 因为服务mtktelephony服务重启后把保存的APN信息清掉了

5W: 为什么服务mtktelephony服务重启后把保存的APN信息清掉了没有触发回调? 因为发生了modem crash重启了服务mtktelephony服务, 丢失了APN状态回调上报

结论:

modem发生crash后网络网卡从up变为linkdown状态, APN实际是断开的状态, 这个场景下没有触发服务mtktelephony服务的APN状态回调上报给上层, 上层APP没有收到APN回调状态通知, 就没有重新建立起APN拨号, TBOX就没有登录TSP后台。

[illegible]

```

37736 20251025 11:39:53.391 eth[2723] [2723] info [monitor_api.cpp:258] eth in AsyncReadyCb
37737 20251025 11:39:53.391 eth[2723] [2723] info [monitor_api.cpp:270] MonitorSendHeartbeat success(eth)
37738 20251025 11:39:53.460 gwm.mcu[2716] [2879] info [mcu_main.cpp:781] topic:101, len:290
37739 20251025 11:39:53.460 gwm.mcu[2716] [2849] info [mcu_main.cpp:965] AppList = 1
37740 20251025 11:39:53.476 gwm.gbServer[2724] [2993] info [gb_report_task.cpp:2361] gb data collection begins.
37741 20251025 11:39:53.568 gwm.mcu[2716] [2879] info [mcu_main.cpp:781] topic:101, len:165
37742 20251025 11:39:53.568 gwm.mcu[2716] [2849] info [mcu_main.cpp:965] AppList = 1
37743 20251025 11:39:53.676 gwm.mcu[2716] [2879] info [mcu_main.cpp:781] topic:101, len:234
37744 20251025 11:39:53.676 gwm.mcu[2716] [2849] info [mcu_main.cpp:965] AppList = 1
37745 20251025 11:39:53.714 gwm.net[2721] [2721] info [net_gdbus_service.cpp:644] Get apn2 dial result [1].
37746 20251025 11:39:53.718 gwm.net[2721] [2721] info [net_gdbus_service.cpp:663] Get sim card[b01] network type[6].
37747 20251025 11:39:53.719 gwm.net[2721] [2721] info [net_gdbus_service.cpp:763] Get sim card[b01] issn[69].
37748 20251025 11:39:53.785 gwm.mcu[2716] [2879] info [mcu_main.cpp:781] topic:101, len:234
37749 20251025 11:39:53.786 gwm.mcu[2716] [2849] info [mcu_main.cpp:965] AppList = 1
37750 20251025 11:39:53.790 eth[2723] [2752] info [eth_main.cpp:530] in RtkStatusCb:3001
37751 20251025 11:39:53.791 gwm.mcu[2716] [2853] info [mcu_main.cpp:781] topic:482, len:15
37752 20251025 11:39:53.791 gwm.mcu[2716] [2849] info [mcu_main.cpp:965] AppList = 1
37753 20251025 11:39:53.792 gwm.gnss[2718] [3130] info [gnss_data_parse.cpp:1462] rtkStatusCode.StatusCode is 3001
37754 20251025 11:39:53.848 gwm.net[2721] [3085] error [dns_resolver.cpp:70] recvfrom failed:Resource temporarily unavailable
37755 20251025 11:39:53.848 gwm.net[2721] [3085] error [net_gdbus_service.cpp:983] DnsGetHostName failed for domain[tbgdm.gwmccloudcn.com]
37756 20251025 11:39:53.849 gwm.tsp[2726] [3288] info [net_api.cpp:904] NetQueryIPv4Addr : ipaddr is [].
37757 20251025 11:39:53.849 gwm.tsp[2726] [3288] error [tsp_socket.cpp:452] call netQueryIPv4Addr errorCode=-1
37758 20251025 11:39:53.850 gwm.net[2721] [2721] info [net_gdbus_service.cpp:695] Set Connection State by Gdbus, profileIndex= [2], state=[1].
37759 20251025 11:39:53.850 gwm.net[2721] [3083] info [net_main.cpp:545] Login TSP failed.
37760 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_socket.cpp:705] close pipe
37761 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_socket.cpp:717] come to SSL close function!
37762 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_socket.cpp:726] close success
37763 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_socket.cpp:738] SSL free success
37764 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_socket.cpp:747] SslCtxRelease success
37765 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_login.cpp:1266] loginSocket Open error
37766 20251025 11:39:53.851 gwm.tsp[2726] [3288] info [tsp_login.cpp:107] SetCurrentUseIpSource is [0]
37767 20251025 11:39:53.851 gwm.tsp[2726] [3288] error [tsp_login.cpp:1938] SocketConnecting failure

```

2025-10-25日19:37:25之后从tsp进程侧看私网上发送任何消息平台不回复，之后重连把域名发给net模块，net模块解析域名也失败。

- 问题分析
- 分析日志：Tbox log

```
Time:20251025113656 cmd: ifconfig
ccmni1: flags=193<UP,RUNNING,NOARP> mtu 1500
    inet6 fe80::1870:adef:bc7c:5396 prefixlen 64 scopeid 0x20<link>
    inet6 240e:540:5d12:2194:1870:adef:bc7c:5396 prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 26 bytes 2288 (2.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1 bytes 48 (48.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni2: flags=193<UP,RUNNING,NOARP> mtu 1500
    inet6 fe80::1870:adf0:4b99:94dc prefixlen 64 scopeid 0x20<link>
    inet6 240e:540:5d16:5b6:1870:adf0:4b99:94dc prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 25 bytes 2200 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1 bytes 48 (48.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni3: flags=193<UP,RUNNING,NOARP> mtu 1500
    inet 10.43.30.194 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 3910836 bytes 3281739058 (3.0 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2959376 bytes 452938994 (431.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni4: flags=193<UP,RUNNING,NOARP> mtu 1500
    inet 20.0.0.20 netmask 255.255.255.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 139670 bytes 12487284 (11.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 265633 bytes 193952753 (184.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni6: flags=193<UP,RUNNING,NOARP> mtu 1500
    inet 10.43.21.149 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 12394852 bytes 621911047 (593.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28193584 bytes 38651966595 (35.9 GiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
Time:20251025113728 cmd: ifconfig
ccmni1: flags=129<UP,NOARP> mtu 1500
    inet6 fe80::1870:adef:bc7c:5396 prefixlen 64 scopeid 0x20<link>
    inet6 240e:540:5d12:2194:1870:adef:bc7c:5396 prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 26 bytes 2288 (2.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1 bytes 48 (48.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni2: flags=129<UP,NOARP> mtu 1500
    inet6 fe80::1870:adf0:4b99:94dc prefixlen 64 scopeid 0x20<link>
    inet6 240e:540:5d16:5b6:1870:adf0:4b99:94dc prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 25 bytes 2200 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1 bytes 48 (48.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni3: flags=129<UP,NOARP> mtu 1500
    inet 10.43.30.194 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 3910993 bytes 3281783551 (3.0 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2959560 bytes 452958348 (431.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni4: flags=129<UP,NOARP> mtu 1500
    inet 20.0.0.20 netmask 255.255.255.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 139691 bytes 12488988 (11.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 265663 bytes 193977178 (184.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni6: flags=129<UP,NOARP> mtu 1500
    inet 10.43.21.149 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 12407398 bytes 622474692 (593.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28222261 bytes 38691171972 (36.0 GiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

从tbox ifconfig查看网卡的信息来看，2025-10-25日19:37:25之后ccmni从up，running状态变为linkdown状态，变为了假网卡，实际APN是断开的。

 /mnt/sdcard/log/mtklog/aee exp/db.00.EE,External (EE),0,0,99,/data/core/,1,modem,,Sat Oct 25 11:37:30 UTC 2025,1,1646447

查看db_history文件有modem crash的历史记录，发生了modem crash了，服务mtktelephonyservice会重启，重启之后APN信息是没有保留的，也没有回调上报。

● 问题分析

- 分析日志: Tbox log

```
0-28 02:09:54.295283 2715 I power: 0x7fb8764-0x ( 2715, 2890),[MAIN],[4],[Tag]ML_Data[TAG]: [ML_SetUnsolResponseFilter] get mdlog state 0, get filter =
0-28 02:09:54.295403 2715 E power: 0x7fb8764-0x ( 2715, 2890),[MAIN],[6],[Tag]ML_MtkLayerAidlClient[TAG]: ITeleService - connect service tele_slot1 suc
0-28 02:09:54.295497 2609 I gsyslogd: 0x5573ed2-0x gwm_syslogdv2.c:860 signal:40 received
0-28 02:09:54.295562 489685 I binder:489685_3: 0x7fa9004-0x (489685, 489688),[RADIO],[4],[Tag]TELE_SERVICE[TAG]: setUnsolResponseFilter different filter,
0-28 02:09:54.295633 489685 I binder:489685_3: 0x7fa9004-0x (489685, 489688),[RADIO],[4],[Tag]TELE_SERVICE[TAG]: setUnsolicitedResponseFilter filter 0
0-28 02:09:54.295822 489685 I binder:489685_3: 0x7fa9004-0x (489685, 489691),[RADIO],[4],[Tag]RfxRootController[-1][TAG]: processMessage() msg = [type=REQ
0-28 02:09:54.296139 489685 I binder:489685_3: 0x7fa9004-0x (489685, 489691),[RADIO],[4],[Tag]RfxDispatchThread[TAG]: addMessageToPendingQueue pRequest =
0-28 02:09:54.296647 2721 W binder:2721_4: 0x7fb01fd-0x [2721] [HalDataCallStopEx:1292] [NOBO-MTK-BSP-DATA-TBOX] CALL HalDataCallStopEx Begin
```

1.APN1和APN3两路公网为什么能恢复正常。因为中间进入了休眠唤醒，休眠过程中有断开APN1和APN3，退出休眠有重新拨号APN1和APN3，这个时候就会触发APN状态的回调上去。退出休眠重新对APN1和APN3拨号后，这两路APN就恢复正常了，公网就能正常上网了。

2.私网APN2为什么没有恢复？因为APN2私网在进入休眠唤醒下也不会触发断开和重建，期间没有APN状态的变化通知，APP就没有发起重建APN的动作。

问题分析

问题场景模拟：手动在tbox敲muxreport 1指令模拟问题场景

1.muxreport 1重启modem，可模拟出ccmni网卡变为linkdown，从ifconfig看还存在这些假网卡，实际是要down掉。

```
root@auto8676p164uostbox:/# muxreport 1
Invoke mux_app mechanism with case=1
root@auto8676p164uostbox:/# ifconfig
bridge6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.16.13.54 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 196 (196.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bridge11: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.16.13.55 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 6 bytes 596 (596.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bridge13: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.13.56 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 38215 bytes 2409810 (2.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2698 bytes 114144 (111.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni1: flags=129<UP,NOARP> mtu 1500
    inet6 240e:57e:30e0:23d:1872:e5d2:4894:b87d prefixlen 64 scopeid 0x0<global>
    inet6 fe80::1872:e5d2:4894:b87d prefixlen 64 scopeid 0x20<link>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 3 bytes 264 (264.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 96 (96.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni2: flags=129<UP,NOARP> mtu 1500
    inet6 fe80::1872:e5d2:a380:b7dd prefixlen 64 scopeid 0x20<link>
    inet6 240e:57e:30e0:23d:1872:e5d2:a380:b7dd prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 4 bytes 352 (352.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 96 (96.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni3: flags=129<UP,NOARP> mtu 1500
    inet 10.80.147.183 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 31703 bytes 3260374 (3.1 MiB)
```

2.查看服务mtktelephonyservice重启后是否有APN状态回调上报，发现是和问题场景一样，没有触发回调上报。

```
//模拟modem crash异常
2020-01-01 00:03:00.817113 2235 I ccci_mdinit: MD1 set status: mtk.mdl.status=exception
//ccmini相关网卡都down掉了，APN实际是断开的
2020-01-01 00:03:00.829448 1638 I mtknetworkmanag: 0x7fad644-0x ( 1638, 1650),[MAIN],[4],[Tag]MtkNetMgr[TAG]: netd_netlinkHandle 600 Iface linkstate ccmni1 down
2020-01-01 00:03:00.837921 1638 I mtknetworkmanag: 0x7fad644-0x ( 1638, 1650),[MAIN],[4],[Tag]MtkNetMgr[TAG]: netd_netlinkHandle 600 Iface linkstate ccmni2 down
2020-01-01 00:03:00.840438 1638 I mtknetworkmanag: 0x7fad644-0x ( 1638, 1650),[MAIN],[4],[Tag]MtkNetMgr[TAG]: netd_netlinkHandle 600 Iface linkstate ccmni3 down
2020-01-01 00:03:00.840677 1638 I mtknetworkmanag: 0x7fad644-0x ( 1638, 1650),[MAIN],[4],[Tag]MtkNetMgr[TAG]: netd_netlinkHandle 600 Iface linkstate ccmni4 down
2020-01-01 00:03:00.840903 1638 I mtknetworkmanag: 0x7fad644-0x ( 1638, 1650),[MAIN],[4],[Tag]MtkNetMgr[TAG]: netd_netlinkHandle 600 Iface linkstate ccmni5 down
2020-01-01 00:03:01.304068 2748 I binder:2748_4: 0x7fb2044-0x ( 2748, 2974),[MAIN],[4],[Tag]ML_Data[TAG]: modem status[ret=6]:mdl:5
2020-01-01 00:03:15.164206 2235 I ccci_mdinit: MD1 set status: mtk.mdl.status=ready
2020-01-01 00:03:15.362536 2748 I binder:2748_4: 0x7fb2044-0x ( 2748, 2974),[MAIN],[4],[Tag]ML_Data[TAG]: modem status[ret=6]:mdl:4
2020-01-01 00:03:16.556895 3969 I mtkfusionrild: 0x7f7f944-0x ( 3969, 3969),[RADIO],[4],[Tag]RILD[TAG]: RIL_Init RIL_register completed
2020-01-01 00:03:16.769756 2759 E tprofiler: 0x7fbd9e4-0x ( 2759, 3298),[MAIN],[6],[Tag]ML_MtkLayerAidlClient[TAG]: The radio tele_slot1 service has died!
2020-01-01 00:03:16.769785 2759 I tprofiler: 0x7fbd9e4-0x ( 2759, 3298),[MAIN],[4],[Tag]ML_MtkLayerAidlClient[TAG]: TeleService destroy, set callback function as null
2020-01-01 00:03:16.769798 2759 E tprofiler: 0x7fbd9e4-0x ( 2759, 3298),[MAIN],[6],[Tag]ML_MtkLayerAidlClient[TAG]: destroy,311 tele setResponseFunctions failed.
2020-01-01 00:03:16.769859 2759 I tprofiler: 0x7fbd9e4-0x ( 2759, 3298),[MAIN],[4],[Tag]ML_MtkLayerAidlClient[TAG]: TeleService destroy, set callback function as null
//mtktelephonyservice 重启
2020-01-01 00:03:16.847003 4049 D mtktelephonyser: 0x7f9c5d4-0x ( 4049, 4049),[RADIO],[3],[Tag]MtkTelephonyService[TAG]: **MtkTelephonyService(yocto-v1.0-2024/12/18) Daemon Started**
2020-01-01 00:03:17.781257 2755 D binder:2755_3: 0x7fa4cf4-0x ( 2755, 4036),[MAIN],[3],[Tag]ML_MtkLayerAidlClient[TAG]: MtkLayerAidlClient init
//mtktelephonyservice 重启后也没有相应的回调上报
2020-01-01 00:03:17.793625 2759 D tprofiler: 0x7fbd9e4-0x ( 2759, 4031),[MAIN],[3],[Tag]ML_MtkLayerAidlClient[TAG]: MtkLayerAidlClient init
```

问题原因

方案：modem crash异常，触发服务mtktelephonyervice重启丢失了APN状态回调上报。

1.服务mtktelephonyervice重启要把APN断开的状态回调上报给上层。

2.断开APN需要把ccmini网卡能够down掉。

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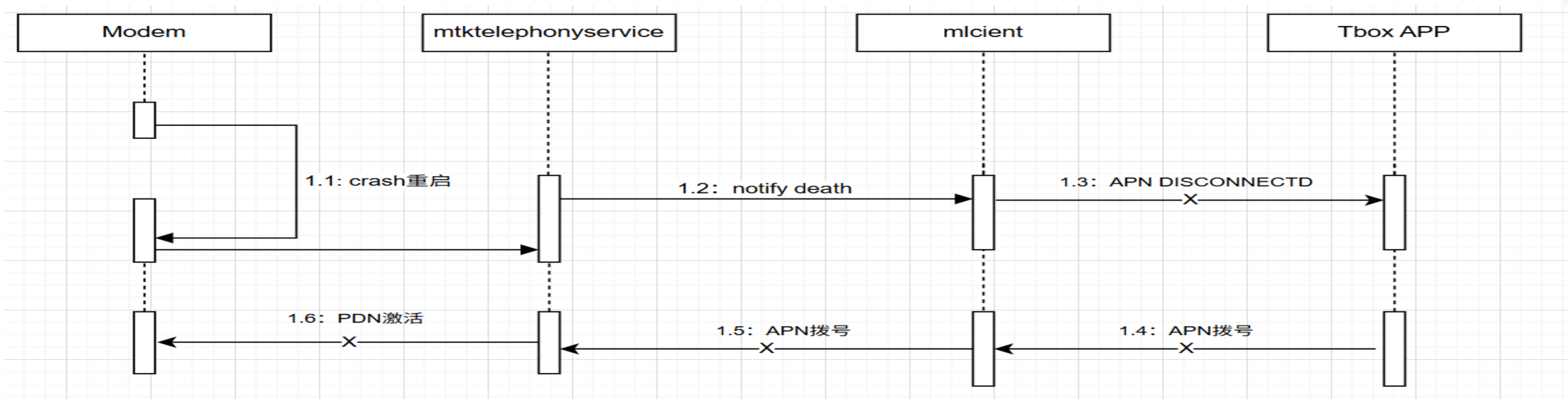
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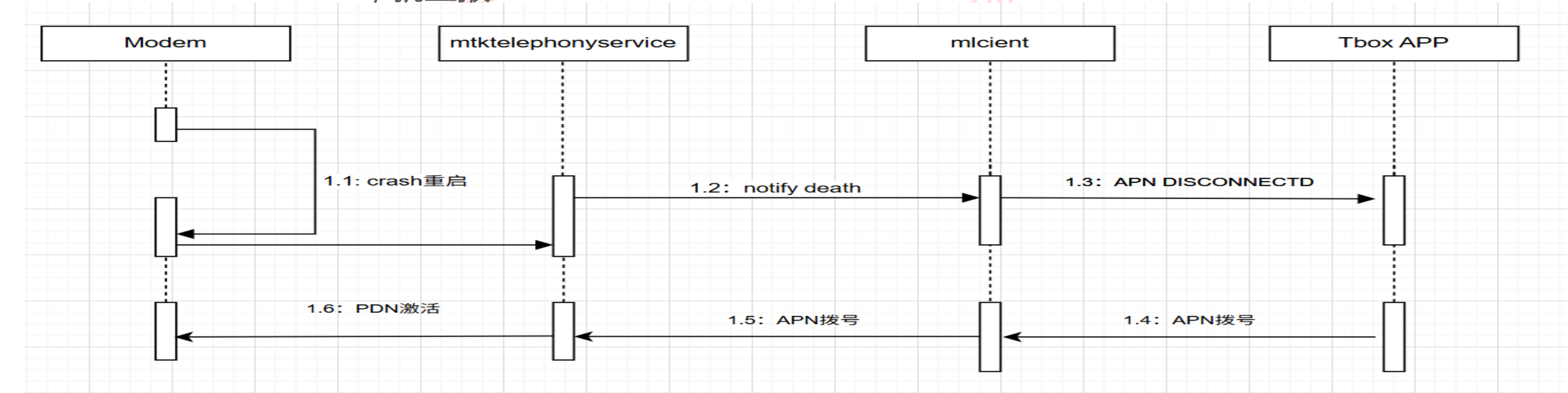
解决方案

1.服务mtktelephonyervice重启要把APN断开的状态回调上报给上层.

异常逻辑：丢失APN断开回调，缺少了1.3的回调上报，导致后续1.4--1.6没有正常激活APN。



修正逻辑：增加1.3APN状态回调上报，mlclient 保存PDN 状态，监测telephonyervice death 时上报，如果PDN 状态不是PDN DISCONNECTED，就上报PDN DISCONNECTED



解决方案

2.断开APN需要把ccmini网卡能够down掉

异常逻辑：网卡通过ifconfig还能看到，但网卡非正常running态，是linkdown状态。

```
root@auto8676p164uostbox:/# muxreport 1
Invoke mux app mechanism with cases1
root@auto8676p164uostbox:/# ifconfig
bridge6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.16.13.54 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 196 (196.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bridge11: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.16.13.55 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 596 (596.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bridge13: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.13.56 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::c8ca:caff:fe23:304 prefixlen 64 scopeid 0x20<link>
    ether ca:ca:ca:23:03:04 txqueuelen 1000 (Ethernet)
    RX packets 38215 bytes 2409810 (2.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2698 bytes 114144 (111.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni1: flags=129<UP,NOARP> mtu 1500
    inet6 240e:57e:30c0:1e9f:1872:e5d2:4894:b87d prefixlen 64 scopeid 0x0<global>
    inet6 fe80::1872:e5d2:4894:b87d prefixlen 64 scopeid 0x20<link>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 3 bytes 264 (264.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 96 (96.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni2: flags=129<UP,NOARP> mtu 1500
    inet6 fe80::1872:e5d2:a380:b7dd prefixlen 64 scopeid 0x20<link>
    inet6 240e:57e:30c0:23d:1872:e5d2:a380:b7dd prefixlen 64 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 4 bytes 352 (352.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 96 (96.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ccmni3: flags=129<UP,NOARP> mtu 1500
    inet 10.80.147.183 netmask 255.0.0.0
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 31703 bytes 3260374 (3.1 MiB)
```

修正逻辑：监听网卡的netlink消息通知，当监听到ccmni网卡的down消息后，通过ifconfig指令把网卡down掉，ifconfig看不到。

```
enp0s0f5: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::a0eb:2ff:fe99:931d prefixlen 64 scopeid 0x20<link>
    ether a2:eb:02:99:93:1d txqueuelen 1000 (Ethernet)
    RX packets 11714 bytes 2482295 (2.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2325 bytes 232914 (227.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s0f5.2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.2.97 netmask 255.255.255.0 broadcast 172.16.2.255
    inet6 fe80::a0eb:2ff:fe99:931d prefixlen 64 scopeid 0x20<link>
    ether 02:47:57:4d:00:97 txqueuelen 1000 (Ethernet)
    RX packets 1736 bytes 311612 (304.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1088 bytes 120893 (118.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s0f5.3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.3.97 netmask 255.255.255.0 broadcast 172.16.3.255
    inet6 fe80::a0eb:2ff:fe99:931d prefixlen 64 scopeid 0x20<link>
    ether 02:47:57:4d:00:97 txqueuelen 1000 (Ethernet)
    RX packets 1761 bytes 303564 (296.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 810 bytes 70595 (68.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s0f5.4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.4.97 netmask 255.255.255.0 broadcast 172.16.4.255
    inet6 fe80::a0eb:2ff:fe99:931d prefixlen 64 scopeid 0x20<link>
    ether 02:47:57:4d:00:97 txqueuelen 1000 (Ethernet)
    RX packets 1141 bytes 247645 (241.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 1976 (1.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

修改代码

1.服务mtktelephonyservice重启把APN断开的状态回调上报给上层

358276 [CUXV5-39986]: fix modem crash, loss apn state callback tp app ☒ src/TeleDataIndication.cpp ☐

Base

110 }
111 return v;
112 }
113 TeleDataIndication::TeleDataIndication(int32_t slot_id) {
114 mSlotId = slot_id;
115 }
116 }
117 }
118 }
119 }
120 }
121 }
122 }
123 }
124 }
125 }
126 }
127 }
128 }
129 }
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155 }
156 }
157 }
158 }
159 }

110 }
111 return v;
112 }
113 TeleDataIndication::TeleDataIndication(int32_t slot_id) {
114 mSlotId = slot_id;
115 }
116 }
117 }
118 }
119 }
120 }
121 }
122 }
123 }
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158 }
159 }

+16 common lines

141 }
142 }
143 }
144 }
145 }
146 }
147 }
148 }
149 }
150 }
151 }
152 }
153 }
154 }
155 }
156 }
157 }
158 }
159 }

+10

146 }
147 }
148 }
149 }
150 }
151 }
152 }
153 }
154 }
155 }
156 }
157 }
158 }
159 }

file

file

+127 common lines

+10

128 }
129 }
130 void* link_thread(void* arg) {
131 //sleep(RETRY_INTERVAL); // Sleep for RETRY_INTERVAL seconds.
132 }
133 MtkLayerAidIClient* instance = MtkLayerAidIClient::getInstance();
134 if (instance == NULL) {
135 ALOGE("MtkLayerAidIClient getInstance return null");
136 return nullptr;
137 }
138 }
139 }
140 }
141 }
142 }
143 }
144 }
145 }
146 }
147 }

128 }
129 }
130 void* link_thread(void* arg) {
131 //sleep(RETRY_INTERVAL); // Sleep for RETRY_INTERVAL seconds.
132 }
133 MtkLayerAidIClient* instance = MtkLayerAidIClient::getInstance();
134 if (instance == NULL) {
135 ALOGE("MtkLayerAidIClient getInstance return null");
136 return nullptr;
137 }
138 }
139 }
140 }
141 }
142 }
143 }
144 }
145 }
146 }
147 }

+375 common lines

147 }
148 }
149 }
150 }
151 }
152 }
153 }
154 }
155 }

+10

147 }
148 }
149 }
150 }
151 }
152 }
153 }
154 }

修改代码

2.断开APN需要把ccmini网卡能够down掉。

```
685     } else if (strstr(msg, "vnet-vccmn1")) {
686         if (mPrivConn[0]->pdn_state == PDN_CONNECTED)
687             mPrivConn[0]->configIPv6(PDN_CONNECTED);
688     } else if (strstr(msg, "vnet-vccmn2")) {
689         if (mPrivConn[1]->pdn_state == PDN_CONNECTED)
690             mPrivConn[1]->configIPv6(PDN_CONNECTED);
691     } else if (strstr(msg, "vnet-vccmn3")) {
692         if (mPrivConn[2]->pdn_state == PDN_CONNECTED)
693             mPrivConn[2]->configIPv6(PDN_CONNECTED);
694     }
695 }
696
697 #ifdef MTK_MBRAIN_SUPPORT
698 if (strstr(msg, "Iface linkstate")) {
699     char *rest = strdup(msg);
700     if (rest == NULL) {
701         ALOGW("%s:%d: strdup! failed.", __func__, __LINE__);
702         return;
703     }
704     char* ifName = getNthToken(rest, 4 /* interface */);
705
706     if (strstr(msg, "up")) {
707         notifyInterfaceChanged(ifName, true);
708     } else if (strstr(msg, "down")) {
709         notifyInterfaceChanged(ifName, false);
710         free(rest);
711     }
712 }
713 #endif
714 }
715
716 void NetworkManager::handleNetlinkMessageforConnLinkDown(const char* msg) {
717     int ret = 0;
718     in_addr_t myaddr = 0;
719     unsigned flags = 0;
720     sp=Connection= targetConn = NULL;
721     char *rest = strdup(msg);
722     if (rest == NULL) {
723         NM_LOGE("%s:%d: strdup failed.", __func__, __LINE__);
724         return;
725     }
726     char* interface = getNthToken(rest, 4 /*interface name*/);
727     if (interface == NULL) {
728         NM_LOGE("%s:%d: getNthToken failed.", __func__, __LINE__);
729         free(rest);
730         return;
731     }
732     if (mPubConn->pdn_state == PDN_CONNECTED
733         && strcmp(interface, mPubConn->iface.c_str()) == 0) {
734         targetConn = mPubConn;
735     } else {
736         for (unsigned i = 0; i < (MAX_PDN_NUMBER - 1); i++) {
737             if (mPrivConn[i]->pdn_state == PDN_CONNECTED
738                 && strcmp(interface, mPrivConn[i]->iface.c_str()) == 0) {
739                 targetConn = mPrivConn[i];
740             }
741         }
742     }
743     if (targetConn) {
744         sp=SetupDataCallResult= dataCallResult =
745             new SetupDataCallResult(targetConn->conn_type);
746         if (dataCallResult == nullptr) {
747             NM_LOGE("new SetupDataCallResult fail\n");
748             free(rest);
749             return;
750         }
751         dataCallResult->pdnState = PDN_IDLE;
752         onDataCall(dataCallResult.get());
753     }
754     ifc_init();
755     ifc_get_info(interface, &myaddr, NULL, &flags);
756     if (flags & IFF_UP) {
757         ret = ifc_down(interface);
758         if (ret) {
759             NM_LOGE("Set down flag on %s: %d\n", interface, ret);
760         } else {
761             NM_LOGE("Set down flag on %s successfully\n", interface);
762         }
763     }
764     ifc_close();
765     if (myaddr != 0) {
766         ret = ifc_clear_addresses(interface);
767         if (ret) {
768             NM_LOGE("Clearing all IP addresses on %s: %d\n", interface, ret);
769         } else {
770             NM_LOGE("Interface %s IP addresses cleared\n", interface);
771         }
772     }
773     free(rest);
774 }
```

```
+10
686     } else if (strstr(msg, "vnet-vccmn1")) {
687         if (mPrivConn[0]->pdn_state == PDN_CONNECTED)
688             mPrivConn[0]->configIPv6(PDN_CONNECTED);
689     } else if (strstr(msg, "vnet-vccmn2")) {
690         if (mPrivConn[1]->pdn_state == PDN_CONNECTED)
691             mPrivConn[1]->configIPv6(PDN_CONNECTED);
692     } else if (strstr(msg, "vnet-vccmn3")) {
693         if (mPrivConn[2]->pdn_state == PDN_CONNECTED)
694             mPrivConn[2]->configIPv6(PDN_CONNECTED);
695     }
696 }
697
698 // error handle for rild or modem crash ?
699 // deconfigNetwork if PDN is CONNECTED
700 handleNetlinkMessageforConnLinkDown(msg);
701
702 #ifdef MTK_MBRAIN_SUPPORT
703 if (strstr(msg, "Iface linkstate")) {
704     char *rest = strdup(msg);
705     if (rest == NULL) {
706         ALOGW("%s:%d: strdup! failed.", __func__, __LINE__);
707         return;
708     }
709     char* ifName = getNthToken(rest, 4 /* interface */);
710
711     if (strstr(msg, "up")) {
712         notifyInterfaceChanged(ifName, true);
713     } else if (strstr(msg, "down")) {
714         notifyInterfaceChanged(ifName, false);
715         free(rest);
716     }
717 }
718 #endif
719 }
```

```
+5 common lines
715     if (strstr(msg, "up")) {
716         notifyInterfaceChanged(ifName, true);
717     } else if (strstr(msg, "down")) {
718         notifyInterfaceChanged(ifName, false);
719         free(rest);
720     }
721 }
722 #endif
723 }
```

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压测结果

- 通过模拟问题场景，muxreport 1重启modem，持续压测验证中。
- 10.31：模拟验证超40次；
 - 1.3路APN都能正常把断开状态callback上去后，3路都正常重拨了。
 - 2.ping公网正常，网络可正常访问。
 - 3.ping私网正常，私网网络可正常访问。

```
root@auto8676p164uostbox:/# ifconfig | grep ccmni
ccmni1: flags=193<UP,RUNNING,NOARP> mtu 1500
ccmni2: flags=193<UP,RUNNING,NOARP> mtu 1500
ccmni3: flags=193<UP,RUNNING,NOARP> mtu 1500
ccmni4: flags=193<UP,RUNNING,NOARP> mtu 1500
ccmni5: flags=193<UP,RUNNING,NOARP> mtu 1500
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/# ping www.baidu.com
PING www.baidu.com (183.2.172.17) 56(84) bytes of data.
64 bytes from 183.2.172.17: icmp_seq=1 ttl=52 time=43.5 ms
64 bytes from 183.2.172.17: icmp_seq=2 ttl=52 time=39.3 ms
64 bytes from 183.2.172.17: icmp_seq=3 ttl=52 time=42.1 ms
^C
--- www.baidu.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 39.319/41.636/43.523/1.743 ms
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/# ping -I ccmni4 10.184.130.2
PING 10.184.130.2 (10.184.130.2) from 20.0.133.188 ccmni4: 56(84) bytes of data.
64 bytes from 10.184.130.2: icmp_seq=1 ttl=58 time=215 ms
64 bytes from 10.184.130.2: icmp_seq=2 ttl=58 time=72.5 ms
^C
--- 10.184.130.2 ping statistics ---
3 packets transmitted, 2 received, 33.3333% packet loss, time 2005ms
rtt min/avg/max/mdev = 72.495/143.968/215.441/71.473 ms
```

公网ping正常

私网ping正常



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